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 Studierichting: Informatica
 Oefeningenreeks 3C nr. 18.16

$$\begin{aligned}
 & \lim_{x \rightarrow 0} \frac{1 - \cos x - \ln \sqrt{1+x^2}}{x^4} \\
 &= \frac{1 - 1 - \ln \sqrt{1+0^2}}{0^4} = \frac{0}{0} \\
 &\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\sin x - \left(\frac{1}{\sqrt{1+x^2}} \cdot \frac{1}{2\sqrt{1+x^2}} \cdot 2x \right)}{4x^3} \\
 &= \lim_{x \rightarrow 0} \frac{\sin x - \frac{x}{1+x^2}}{4x^3} \\
 &= \frac{0 - \frac{0}{1+0}}{0} = \frac{0}{0} \\
 &\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\cos x - \frac{(1+x^2) - x(2x)}{(1+x^2)^2}}{12x^2} \\
 &= \lim_{x \rightarrow 0} \frac{\cos x - \frac{1-x^2}{(1+x^2)^2}}{12x^2} \\
 &= \frac{1 - \frac{1-0}{(1+0)^2}}{0} = \frac{0}{0} \\
 &\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{-\sin x - \left(\frac{-2x(1+x^2)^2 - (1-x^2) \cdot 4x(1+x^2)}{(1+x^2)^4} \right)}{24x} \\
 &= \lim_{x \rightarrow 0} \frac{-\sin x - \left(\frac{(1+x^2) \cdot [-2x(1+x^2) - 4x(1-x^2)]}{(1+x^2)^4} \right)}{24x} \\
 &= \lim_{x \rightarrow 0} \frac{-\sin x - \left(\frac{-2x(1+x^2) - 4x(1-x^2)}{(1+x^2)^3} \right)}{24x} \\
 &= \lim_{x \rightarrow 0} \frac{-\sin x - \left(\frac{2x^3 - 6x}{(1+x^2)^3} \right)}{24x} \\
 &= \frac{-0 - \frac{0-0}{1}}{0} = \frac{0}{0} \\
 &\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{-\cos x - \left(\frac{(6x^2-6)(1+x^2)^3 - (2x^3-6x) \cdot 3(1+x^2)^2 \cdot 2x}{(1+x^2)^6} \right)}{24}
 \end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \frac{-\cos x - \left(\frac{(6x^2 - 6)(1 + x^2) - (2x^3 - 6x) \cdot 6x}{(1 + x^2)^4} \right)}{24} \\
&= \lim_{x \rightarrow 0} \frac{-\cos x - \left(\frac{6x^2 - 6 + 6x^4 - 6x^2 - 12x^4 + 36x^2}{(1 + x^2)^4} \right)}{24} \\
&= \lim_{x \rightarrow 0} \frac{-\cos x - \left(\frac{-6x^4 + 36x^2 - 6}{(1 + x^2)^4} \right)}{24} \\
&= \frac{-1 + 6}{24} = \frac{5}{24}
\end{aligned}$$

References